4.3 Sample Files – Common Table Expressions

Business Question: Which artists are on the best trend to produce more music?

Metric Used: Artist Revenue Growth Trend (with a minimum sales volume threshold)

I used a revenue-growth trend as a proxy metric, comparing each artist's average yearly revenue in an early period (2009–2011) against a recent period (2012–2013), while filtering out artists with too few sales to be statistically meaningful.

SQL Query

sql

```
WITH ArtistYearSales AS (
   SELECT
      ar.ArtistId,
      ar.Name AS ArtistName,
      strftime('%Y', i.InvoiceDate) AS SalesYear,
      COUNT(il.InvoiceLineId) AS TracksSold,
      SUM(il.UnitPrice * il.Quantity) AS YearlyRevenue
   FROM Artist ar
   JOIN Album al ON ar.ArtistId = al.ArtistId
   JOIN Track t ON al.AlbumId = t.AlbumId
   JOIN InvoiceLine il ON t.TrackId = il.TrackId
   JOIN Invoice i ON il.InvoiceId = i.InvoiceId
   GROUP BY ar.ArtistId, ar.Name, SalesYear
),

ArtistTrend AS (
   SELECT
      ArtistId,
```

```
    ArtistName,
    SUM(TracksSold) AS TotalTracksSold,
    AVG(CASE WHEN SalesYear IN ('2009','2010','2011') THEN YearlyRevenue END) AS EarlyAvgRevenue,
    AVG(CASE WHEN SalesYear IN ('2012','2013') THEN YearlyRevenue END) AS RecentAvgRevenue
  FROM ArtistYearSales
  GROUP BY ArtistId, ArtistName
),

ArtistGrowth AS (
  SELECT
    ArtistId,
    ArtistName,
    TotalTracksSold,
    ROUND(EarlyAvgRevenue, 2) AS EarlyAvgRevenue,
    ROUND(RecentAvgRevenue, 2) AS RecentAvgRevenue,
    ROUND(((RecentAvgRevenue - EarlyAvgRevenue) / EarlyAvgRevenue) * 100, 2) AS RevenueGrowthPercent
  FROM ArtistTrend
  WHERE EarlyAvgRevenue IS NOT NULL
    AND RecentAvgRevenue IS NOT NULL
    AND EarlyAvgRevenue > 0
    AND TotalTracksSold >= 5
)

SELECT
```

```
    ArtistName,
    TotalTracksSold,
    EarlyAvgRevenue,
    RecentAvgRevenue,
    RevenueGrowthPercent,
    DENSE_RANK() OVER (ORDER BY RevenueGrowthPercent DESC) AS GrowthRank
FROM ArtistGrowth
ORDER BY RevenueGrowthPercent DESC
LIMIT 10;
```

How it works:

- ArtistYearSales aggregates each artist's tracks sold and revenue per year.
- ArtistTrend compares each artist's average revenue in an early period against a recent period.
- ArtistGrowth calculates the percentage change and filters out artists with fewer than 5 total tracks sold, to avoid small-sample distortion.
- The final SELECT ranks the top 10 artists by revenue growth percentage.

1. Three Reasons This Is a Good Metric

It captures direction, not just size. Comparing average revenue across two time periods shows whether customer demand for an artist is rising or falling, which is closer to a "trend" than a single lifetime snapshot.

The volume threshold improves reliability. Requiring a minimum of 5 tracks sold before an artist is ranked prevents artists with only 1–2 sales from showing artificially extreme growth percentages.

It's modular and easy to audit. Breaking the logic into CTEs (aggregate - compare periods - rank) means each stage of the analysis can be reviewed or adjusted independently, which supports transparent business decision-making.

2. Three Limitations of This Metric

Revenue is a demand proxy, not a production measure. The Chinook database has no release-date data, so I cannot verify that an artist is actually creating more music, only that customers are buying more of their existing catalog.

External factors can distort revenue independent of the artist. Pricing changes, promotions, or a single popular track can inflate a period's sales without reflecting genuine, sustained artistic momentum.

The time window is limited by the dataset. Splitting only 2009-2013 into two short periods restricts how confidently I can distinguish a real multi-year trend from a temporary spike.

3. Additional Data That Would Improve This Metric

Actual release dates and release frequency: knowing how many albums or tracks each artist releases per year would let me measure production directly instead of inferring it from sales.

Streaming and engagement data (plays, saves, playlist adds): this would reflect current audience interest more accurately than purchase transactions alone.

Marketing and cost data: knowing what was spent on promotion for each artist would let me separate genuine growing demand from revenue increases driven by ad spend, supporting a more accurate ROI-based decision.

Closing Note

This metric does not predict which artists *will* produce more music but it identifies artists showing rising customer demand within the available sales data, which the record label could use as one input, alongside other data sources, when deciding where to invest in additional music production.